SIGNAL DESK

Comprehensive Business Plan & Operational Model

Strategic Framework for a Cloud-Based Public Infrastructure Triage & Asset-Liquidation Platform

A high-margin B2G and B2B public infrastructure data ecosystem capturing real-time municipal triage logs, citizen routing, and multi-team field dispatch dynamics to license proprietary operational datasets to frontier AI engineering labs (Anthropic, Google, OpenAI, X/Cursor, Meta).

SEED RUNWAY	YEAR 5 REVENUE	YEAR 5 NET PROFIT	GROSS MARGIN
$100k (18-24 mo)	$17.425M	+$15.260M	89.9%

Executive Summary: High-Margin Infrastructure Data Ecosystem

Commercial deployment of a cloud-based public infrastructure triage & asset-liquidation platform.

SOURCE LAYER

Proprietary Data Ingestion

By capturing real-time municipal triage logs, citizen utility routing patterns, and multi-team field dispatch dynamics, the platform creates an un-scraped, highly proprietary operational dataset unavailable in open public repositories.

HIGH-MARGIN SCALE

Frontier AI Lab Licensing

Beginning in Year 2, processed structural datasets are licensed directly to frontier AI labs (Anthropic, Google, OpenAI, X/Cursor, Meta) for specialized neural networks, agentic workflows, and Reinforcement Learning (RL) training.

RUNWAY SECURITY

Capital Efficient Runtime

A $100,000 initial seed round funds 18 to 24 months of core runtime with a lean $75,000 Year 1 total baseline outflow, enabling proof of unit economics and data refinery scale before raising institutional growth capital.

Financial Feasibility & Seed Runway Allocation ($100,000)

Initial seed round allocation providing 18 to 24 months of core operational runtime.

Sunk Assets (Historical R&D)

Prior software design iterations and foundational engine engineering capitalised as a sunk setup asset.

$30,000

Net Cash Position (Working Capital)

Initial liquid working capital deployed into active operations, cloud hosting, and councillor outreach.

$70,000

Fixed Overhead Core Infrastructure

Baseline cloud server deployments ($500/month) and dedicated Claude API workflow layers ($1,000/month).

$18,000 / yr

Year 1 Total Baseline Outflows

Combined with lean Year 1 Operating Expenses of $57,000 ($45k payroll + $12k other opex) + $18k infrastructure.

$75,000 / yr

Key Runway Takeaway: The $100,000 seed round funds 18 to 24 months of core runtime, providing ample time to build proprietary data assets and secure self-sustaining revenue before follow-on investment.

Market Positioning & Go-To-Market (GTM) Strategy

Circumventing traditional public sector procurement friction via a bottom-up adoption wedge.

STAGE 1 - LOW FRICTION

Bottom-Up Councillor Wedge

Pricing: $10 / month

Direct outreach loops introduce individual ward councillors to dedicated utility tracking dashboards. Allows real-time tracking of community service requests, building organic ward adoption at low friction.

STAGE 2 - WORKFLOW LINK

B2B Service Provider Onboarding

Pricing: $50 / month

Contracted external municipal vendors (plumbing, road maintenance, electrical repair crews) use the interface for structured dispatches, drastically lowering manual verification and inspection overhead.

STAGE 3 - INSTITUTIONAL

Data Asset Liquidation

Pricing: $100k -> $1.0M / package

As infrastructure data deepens, the platform transitions into a high-tier data refinery. Anonymized, end-to-end municipal triage and resolution datasets are licensed to frontier AI labs.

CivicRewards Loyalty Framework

Maintaining reporting volume and verified field data without increasing operational cash burn.

01

Citizen Report

Citizen submits verified infrastructure gap via Telegram mini-app / web.

02

Geolocation & De-dupe

System validates GPS coordinates, identifies duplicate tickets, and maps official refs.

03

Token Allocation

CivicPoints awarded when service gap is independently validated by community resolution.

04

Partner Redemption

Points redeemed for local merchant discounts funded by B2B Advertisers ($1,000/mo).

Validation Engine

Citizens earn loyalty tokens ('CivicPoints') when they log verified public service gaps that are subsequently validated by community resolution engines. This ensures high signal-to-noise ratio in incoming municipal dispatch feeds.

Loop Monetization (Zero Cash Burn)

Points are redeemable for localized value-added discounts and micro-rewards. These incentives are funded directly by Premium B2B Advertisers paying $1,000/month, keeping customer engagement high without creating any direct cash burn for Signal Desk.

Data Refinement & AI Broker Monetization Timeline

Platform data assets increase in valuation as transactional complexity and running history accumulate.

YEAR 2 - QUALIFIED PARTNERSHIPS

$100,000 / package

Target: 1 Broker (micro1)

The platform licenses basic operational logs, citizen routing patterns, and triage histories under pilot data partnership programs.

YEAR 3 - LARGE DATA COLLECTIONS

$500,000 / package

Target: 2 Brokers ($1.0M Total)

Capturing ongoing, multi-team, deep transactional logs and field work histories across 20 partner municipalities.

YEARS 4 & 5 - HIGHLY UNIQUE / PROPRIETARY

$1,000,000 / package

Target: Anthropic, Google, OpenAI, X/ Cursor, Meta

Distinctive operational workflows involving complex human decisions, triage, and escalations. 3 partner municipalities produce 1 distinct dataset/ yr sold across 5 active brokers ($15.0M in Year 5).

5-Year Operational & Accounts Scaling Targets

Target accounts scaling across software seats, partner municipalities, and AI data brokers.

Account Segment	Year 1	Year 2	Year 3	Year 4	Year 5
Ward Councillors ($10/mo)	100	300	800	1,800	3,200
B2B Service Providers ($50/mo)	20	50	120	250	500
Partner Municipalities	3	8	20	45	90
Premium Advertisers ($1k/mo)	3	8	15	30	50
Data Analytics Clients	1	3	7	15	30
AI Data Brokers (Frontier Labs)	**0**	**1 (micro1)**	**2**	**4**	**5 (Tier 1)**

Notes: AI Data Brokers in Year 4 & 5 represent structured dataset partnership contracts with Anthropic, Google, OpenAI, X/Cursor, and Meta.

Table A: 5-Year Consolidated Financial Projection (USD)

Pro-forma income statement modeling exponential revenue growth and 89.9% Year-5 gross margins.

Account Segment	Year 1	Year 2	Year 3	Year 4	Year 5
Gross Annual Revenue	**$30,000**	**$315,000**	**$1,523,000**	**$13,261,000**	**$17,425,000**
Cost of Revenue (COR)	$22,500	$62,100	$186,990	$1,361,520	$1,760,500
Gross Profit	$7,500	$252,900	$1,336,010	$11,899,480	$15,664,500
Gross Margin %	**25.0%**	**80.3%**	**87.7%**	**89.7%**	**89.9%**
Staffing Headcount	3 Staff	4 Staff	6 Staff	9 Staff	13 Staff
Baseline Core Payroll	$45,000	$70,000	$120,000	$200,000	$310,000
Other Operating Expenses	$12,000	$20,000	$35,000	$60,000	$95,000
Total Operating Expenses (Opex)	$57,000	$90,000	$155,000	$260,000	$405,000
Net Operating Profit / (Loss)	**(-$49,500)**	**+$162,900**	**+$1,181,010**	**+$11,639,480**	**+$15,259,500**

Operational Cost Architecture & Lean Delivery Framework

Structural efficiency ensures operating expenses remain below 2.5% of revenue by Year 5.

Cost of Revenue (COR) Architecture

- Baseline Core Infrastructure: $18,000 per year fixed baseline comprising cloud servers ($500/mo) and dedicated Claude API workflow layers ($1,000/mo).

- Variable Platform Usage: Decreases systematically from 15% in Year 1 down to 10% in Year 5 as server efficiency, ticket deduplication, and cache hit rates scale.

- Near-Zero Marginal Data Cost: Once raw incident and dispatch telemetry is captured and linked, dataset compilation and packaging for AI labs requires minimal marginal compute.

- Operating Leverage: Gross margin expands from 25.0% in Year 1 to 89.9% in Year 5 as institutional data broker revenue outpaces operational delivery expenses.

Operating Expense (Opex) Discipline

- Lean Engineering Payroll: Capitalizes on high-caliber South African technical talent ($45k Year 1 -> $310k Year 5 across 13 engineering and operations personnel).

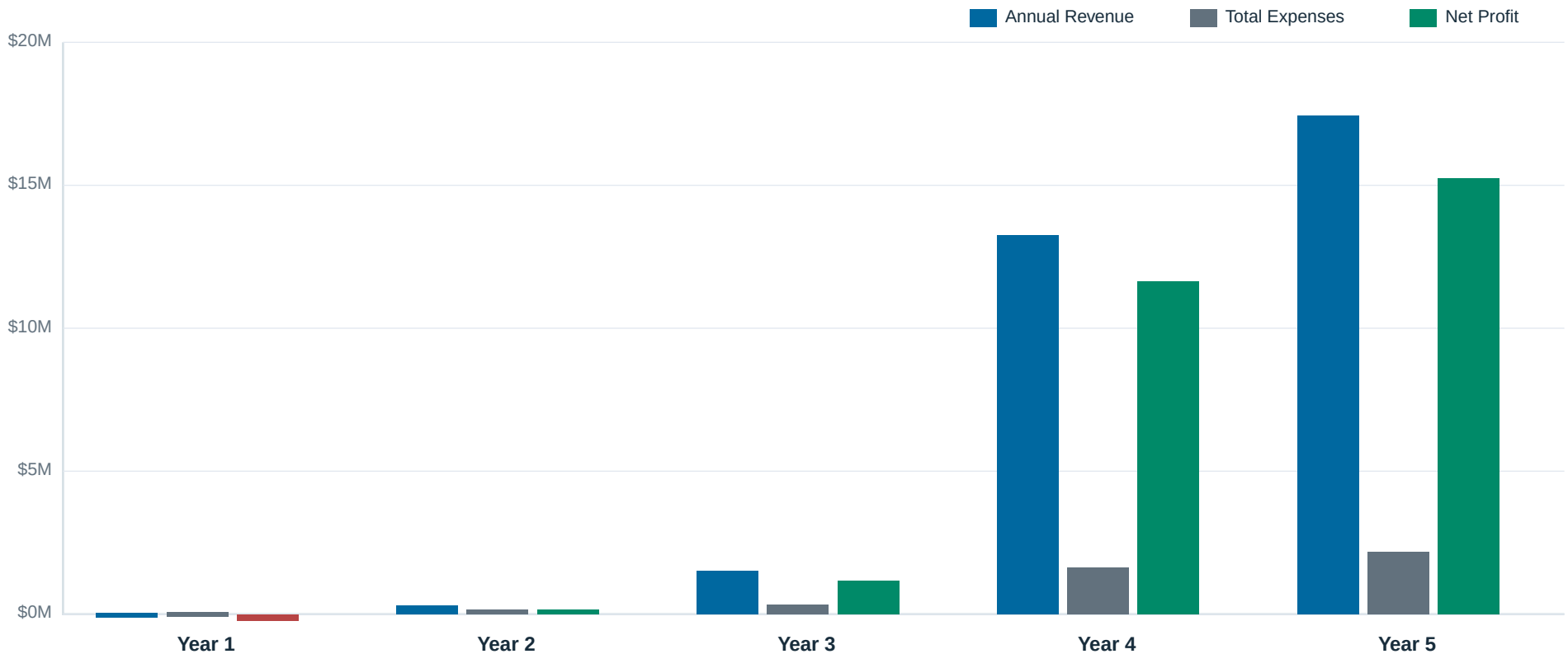
- Below-Market Founder Compensation: Founder salaries maintained strictly below market benchmarks early in the lifecycle to protect liquid runway.

- Strict Travel & Procurement Cap: Zero speculative municipal lobbying or travel; all external development expenses bounded inside the 'Other Opex' envelope ($12k -> $95k).

- Profitable by Year 2: Net operating turnaround generates +$162,900 profit in Year 2, funding internal growth without early dilutive down-rounds.

Chart 1: 5-Year Financial Trajectory & AI Lab Data Scaling

Comparison of Annual Revenue, Total Operational Expenses (COR + Opex), and Net Operating Profit.

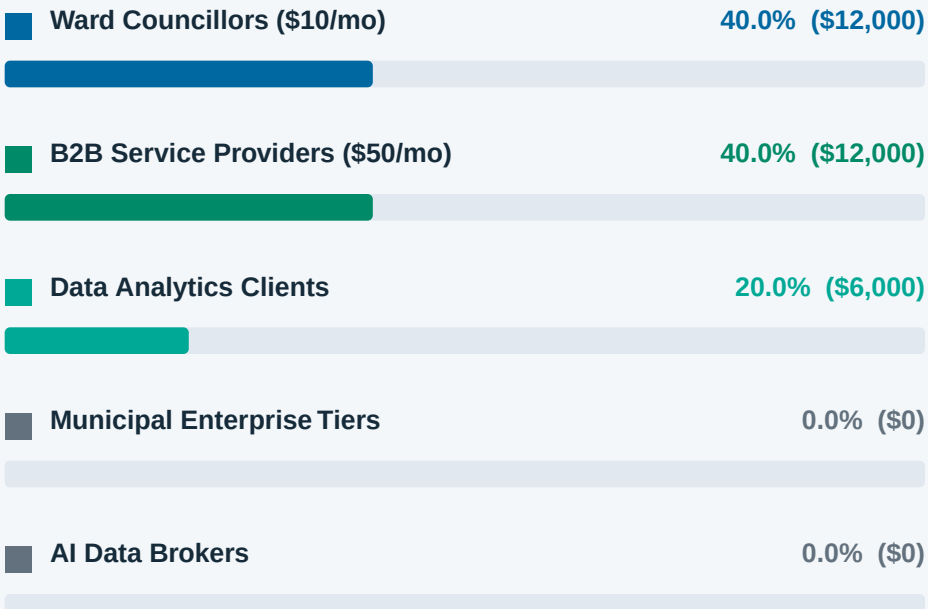


Trajectory Summary: Explosive top-line revenue scaling from $30k (Y1) to $17.425M (Y5) driven by 86.1% AI Data Broker dataset licensing.

Charts 2 & 3: Revenue Stream Evolution (Year 1 vs. Year 5)

Transition from early municipal seat subscriptions to global AI data asset refinery.

Year 1: Direct Software Footprint ($30,000 Total)



Year 5: Global Data Refinery ($17,425,000 Total)

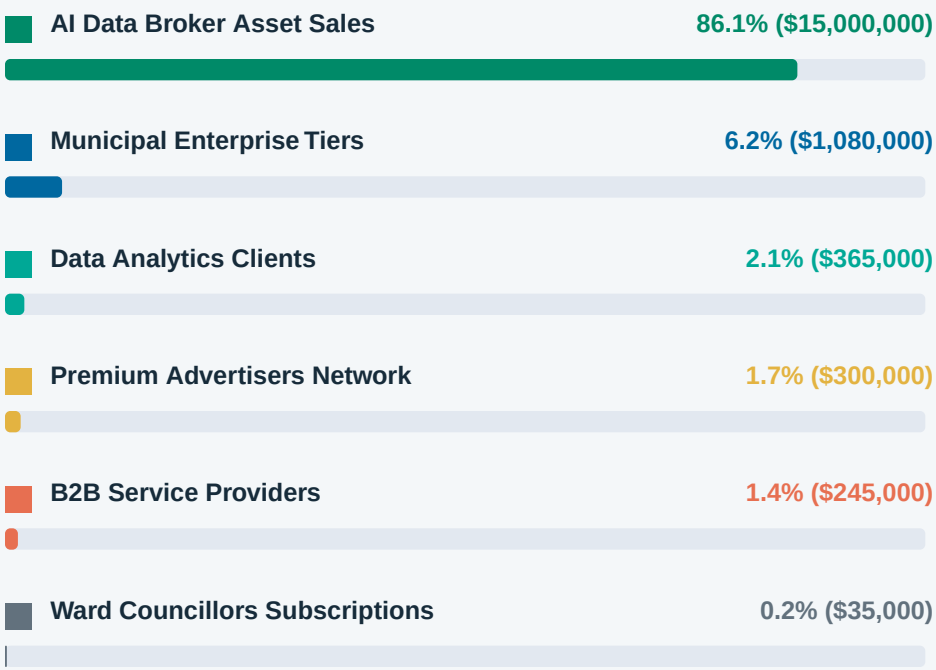
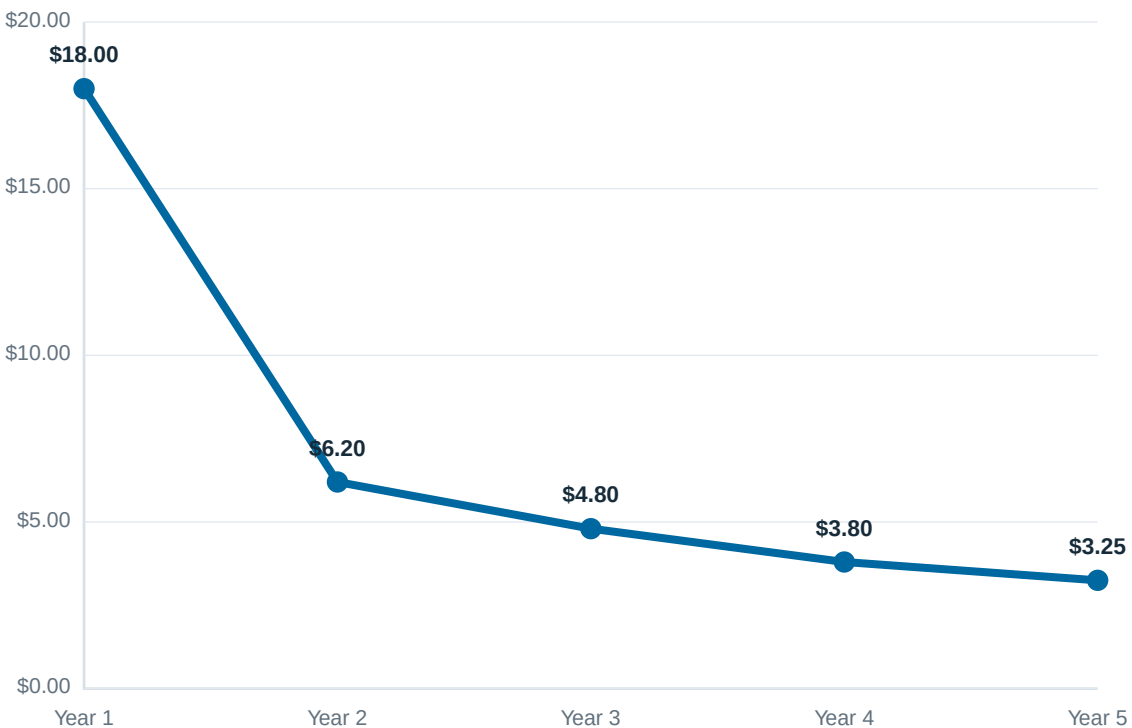


Chart 4: Unit Economics - Cost to Serve (CTS) Decay Curve

Single-ward server costs decrease as system throughput and data monetization scale.



Year 1 CTS ($18.00 / month)

Consists of $3.00 variable AI parsing token cost (150 tickets @ $0.02) plus $15.00 amortized server overhead across the initial cohort of 100 active wards.

Year 2 CTS ($6.20 / month)

Fixed infrastructure amortizes significantly as ward count scales to 300 active wards, while ticket deduplication and caching lower token costs.

Year 5 CTS ($3.25 / month)

Fixed server overhead is virtually 100% amortized by data broker asset licensing. Single-ward contribution margin expands to 83.8%.

Core Unit Economics & Single-Ward Micro Analysis

Evaluating profitability at the individual paying ward footprint reveals structural network efficiency.

Target Ward Composition

1 Councillor + 0.2 Provider

1 local Councillor seat ($10/mo) paired with approximately 0.2 integrated Service Provider access configurations ($10/mo pro-rata value share).

Blended Ward Monthly Revenue

$20.00 / month ($240/yr)

Reliable annualized recurring SaaS subscription yield per active ward footprint before factoring in institutional data licensing.

Variable Cost to Serve (CTS)

$3.00 / month

Based on pilot metrics: $0.02 variable AI token cost per validated incident report x 150 incoming validated tickets per ward monthly.

Server Overhead Amortization

$15.00 / month (Y1)

$18,000/yr hosting + dedicated Claude API connectivity allocated evenly across the initial cohort of 100 active wards.

Net Contribution Margin Expansion

10.0% -> 83.8%

Monthly ward contribution margin expands from $2.00 (10%) in Year 1 to $16.75 (83.8%) in Year 5 as server CTS declines to $3.25.

Cost of Acquisition (CAC)

$45.00 / ward

Direct outreach channels to municipal councillor cohorts achieve full capital payback within 2.25 months (< 1 quarter).

Data Governance, Privacy (POPIA) & Frontier AI Training

Transforming raw civic messages into legally compliant, clean, structured training collections.

PILLAR 1

Privacy & De-Identification

Complete stripping of personal identifiers (PII), telephone numbers, and resident identities in strict compliance with South Africa's Protection of Personal Information Act (POPIA) and global data sovereignty frameworks.

PILLAR 2

Canonical Triage Graph

Normalizing the real-time operational lifecycle: raw report -> automated categorization -> departmental assignment -> contractor dispatch -> physical repair -> community verification.

PILLAR 3

Agentic Training Utility

Frontier AI labs acquire this data to train autonomous multi-agent operational models capable of navigating physical constraints, human escalation loops, and municipal logistics.

Summary & Investment Milestones

Clear execution path from initial seed validation to institutional data scale.

MONTHS 1-12

Phase 1: Seed Validation

- Deploy $100k seed runway ($75k Y1 burn)
- Onboard 100 Ward Councillors ($10/mo)
- Onboard 20 Service Providers ($50/mo)
- Establish $30k Year 1 baseline revenue
- Validate 150 tickets/mo/ward volume
- Prove 2.25-month CAC payback

MONTHS 13-24

Phase 2: Data Commercialization

- Scale to 300 wards & 8 municipalities
- License pilot dataset to micro1 ($100k)
- Achieve net profitability (+$162.9k)
- Launch CivicRewards ad loop ($1k/mo)
- 18-24 month self-funding runway
- Solidify South African operating core

YEARS 3-5

Phase 3: Global AI Refinery

- Expand to 3,200 wards & 90 municipalities
- Deliver datasets to 5 frontier AI labs
- Scale top-line revenue to $17.425M
- Achieve 89.9% margin & +$15.26M profit
- Global standard for municipal triage AI
- Institutional liquidity & regional expansion